

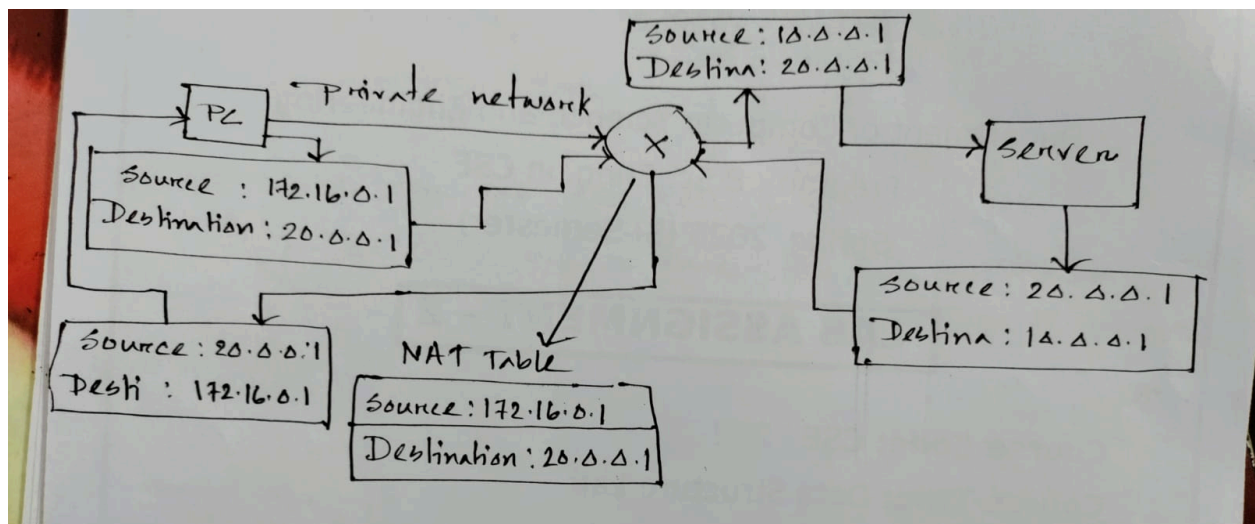
How NAT(Network address Translation) works :

a. Outgoing Request (PC to server)

1. A Device on private network wants to send data in web server in the internet. For example : Source IP : 172.16.0.1, Destination IP : 20.0.0.1
2. when the package reached the router(10.0.0.1), the router replace its private IP address to its IP address. Now Source Address : 10.0.0.1 , Destination Address : 20.0.0.1
3. The router Saves the mapping in NAT/routing table
Private IP - 172.16.0.1 , public IP - 10.0.0.1

b. Incoming response (Server to PC)

1. Server reply to router by sending data back. Now source IP : 20.0.0.1 , Destination IP : 10.0.0.1
2. Now router checks from NAT/routing table to see which local device is request that data. It replaces the destination IP with the original private IP.
3. the packet is successfully reach to the original PC



How PAT(Port Address Translation) works :

1. Suppose PC1 and PC2 both wants to access Web server .

For PC1 : Source IP : 172.16.1.1, Destination IP : 20.0.0.1

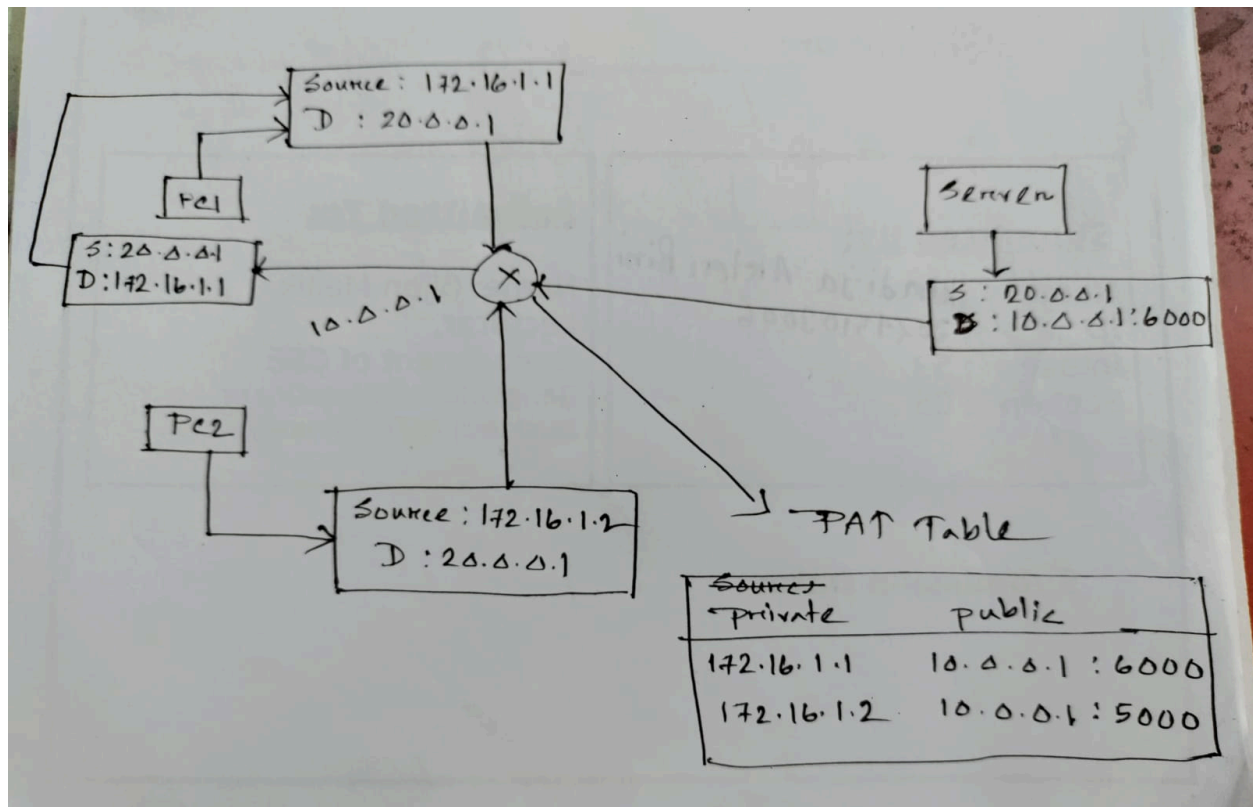
For PC2 : Source IP : 172.16.1.2 Destination IP : 20.0.0.1

2. Both request reach the router(10.0.0.1) .Router replace their private IP with same public IP from router. Just give different port number. Now this information saves the PAT table so that router know which internal device should receive each response.

Private address : 172.16.1.1 -----public address : 10.0.0.1 : 6000

Private address : 172.16.1.2 -----public address : 10.0.0.1 : 5000

3. suppose server response to router by 10.0.0.1 : 6000 IP address. Now source IP : 20.0.0.1 , destination IP : 10.0.0.1 : 6000. Now router checks from PAT table which local device send request for 20.0.0.1 and got the result 172.16.1.1. Then the packet send to the PC1 successfully



Flooting :

Flooding is a routing technology in which the package pass through all possible path (Except from the path it arrives)

controlled Flonding: There are some rules to Prevent duplication and reduce network congestion.

sequence number: All packet has unique sequence number.so router realize if duplicate packet wants to for forward again.

Hop count/TTL : There is a limit of how many router can cross a packet. If the limit is over the packet discarded.

Uncontrolled Flooding : There is no rule for passing, packet through router. So duplication is very common and network congestion can happen.On the other hand, it is very simple and reliable for reaching destination